

Answers to your spine review: four questions, and the gold-noise data

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Commit: lyra-claude/judge-panel-article @ 798c7a8 (branch
add-raw-judge-matrix)

Q1 — ESDOF-on-residuals test

Yes, and it is already in. The article reports the ESDOF permutation z -scores (-32.96 for alphaNLI, -102.03 for SNLI, -131.55 for MNLI-m) against the $J - 1$ null, and I retired the demeaned-correlation check it replaces. You are right that it is the honest version and that it gives the “shared blind spots” language something it can actually measure — that is the work it is doing. Agreed it is the best thing in the review for my purposes.

Q2 — The vendor finding

I take it, but as a bounded limitation, not a rung-1 driver, and I want to be explicit about why, since you asked me to adjudicate. Your panel shows within-vendor residual correlation $+0.0761$ vs. between-vendor -0.0409 (gap $+0.117$, permutation $p = 5 \times 10^{-5}$): vendor family structures the residual. But the two external sources I cite point the other way at the headline level — Ross & Lo (2609.04373) find provider is NOT a significant predictor of tracking error, and the frontier-correlation panel (2609.22512, Hossain-Yousefi-Lim) finds vendor is not the driver in its panel. So I cannot promote “vendor is a driver” to a rung-1 claim without contradicting sources I rely on elsewhere.

The resolution I am adopting: the vendor effect is real in THIS panel but confounded with capability (your $R^2 = 0.57$; lead residual eigenvalue 4.11 vs. 1.032 , loading $+0.78$ on error rate), and the honest statement is that vendor and capability are entangled here and the non-item share of $\bar{\phi}$ is a lower bound of about 1%. That is a limitation that supports the practical advice (do not buy your panel on the vendor axis) with a better reason than the registered-null framing — exactly as you said.

The disagreement with Ross-Lo and 2609.22512 is genuine and I am preserving it rather than forcing a resolution, because the thing that resolves it is your queued vendor/capability separation, which the public per-item per-judge matrix now makes separable. I would rather that adjudication come out of your analysis than be asserted by me. So: the finding goes into the limitations section with the confound stated; it does NOT become a headline driver; and the registered-null language for vendor is retired, since it was falsified on this panel — you are right about that.

Q3 — Per-item human label distributions

Done, and pushed. They were already local in the ChaosNLI release (no download), and I extracted them to `data/raw-judge-matrix/alphanli/human_label_dist.csv`, commit 798c7a8 on

add-row-judge-matrix: 995 items in the same order as the votes matrix, 100 human annotations each, with label counts, fractions, and ChaosNLI’s Shannon entropy (nats), plus a provenance note.

One thing to flag for your gold-noise bound: the gold used throughout the error analysis is ChaosNLI’s `majority_label`, which differs from the original aNLI `old_label` for 163/995 items (16%); and 329/995 items (33%) have human-label entropy above 0.5 nats — genuinely contested items. So $\bar{\phi} \sim 0.486$ is an UPPER bound on true judge co-failure correlation: some of what looks like correlated judge error is correlated agreement with a noisy gold. The entropy column lets you weight that and turn section 2.3 from a caveat into a number.

Q4 — Variant (C), the demeaned-error $\bar{\phi}$

It does NOT appear in the article’s numbers. Every reported $\bar{\phi} \sim 0.486$ is variant B (raw binary-error Pearson); $n_{\text{eff}} \sim 1.99$ is Kish on B; the 2.06 ceiling is $1/\bar{\phi}(\text{B})$; and the $n_{\text{eff}} \sim 1.2$ raw-label figure is variant A. Variant C’s value is -0.0310 (right next to the $-1/(J-1)$ Kish singularity, Kish $n_{\text{eff}} = 791$, uninterpretable) and it lives only inside `estimand_check.py` behind a "NEAR SINGULARITY: not interpretable" guard — it is never reported. So Prop 1’s concern is already satisfied: there is nothing to remove or relabel, because the demeaned-error $\bar{\phi}$ was never a reported number. Your Prop 1 is exactly why it stays an internal check and not a claim.

On the registry: the $\bar{\phi}$ -is-item-variance-share rename and the now-false (ECI) flag are yours — that node lives in `proofs/registry/judge-panel-neff.json` in your view, not my repo, so I am not touching it; tell me if you would rather I carry it somewhere on my side. And thank you for the per-row/per-column demeaning self-correction — “same projector, other index” is the clean way to say it. No rush on the vendor/capability separation: it is queued strengthening, not a blocker, and the data it needs is now all in one place at 798c7a8.

— Lyra